Tumeur ténosynoviale à cellules géantes (TTCG / Synovite villonodulaire) : évaluation IRM et efficacité à long terme des thérapies ciblées

Titre original : Long-term efficacy of imatinib mesylate in patients with advanced Tenosynovial Giant Cell Tumor

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1. Pertinence Clinique & Lien avec l'Agenda Interventionnel (Cochin)

En lien direct avec les IRM de contrôle de cheville/genou pour TTCG / synovite villonodulaire (SVN) sous traitement médical et les suivis post-cryothérapie à Cochin.

2. Synthèse Clinique en Français

Résumé exécutif : Étude multicentrique évaluant la réponse clinique et radiologique (RECIST et score volumétrique TVS en IRM) des tumeurs ténosynoviales à cellules géantes (TTCG / synovite villonodulaire pigmentée diffuse) traitées médicalement lorsqu'une synovectomie chirurgicale complète est impossible ou morbide.

Analyse détaillée : Les formes diffuses de tumeur ténosynoviale à cellules géantes (TTCG, anciennement synovite villonodulaire pigmentée - SVN) de la cheville, du genou ou de la hanche présentent un taux élevé de récidive locale après chirurgie. L'IRM est l'examen clé pour le diagnostic (effet de susceptibilité magnétique 'blooming' lié aux dépôts d'hémosidérine en écho de gradient T2*) et pour le suivi longitudinal sous thérapies médicales (inhibiteurs de CSF1R tels que pexidartinib/vimseltinib, imatinib ou biothérapies). L'évaluation IRM doit combiner la mesure volumétrique synoviale, le degré de rehaussement et l'état chondral articulaire.

3. Protocole Technique & Paramètres Opératoires

Protocole IRM TTCG / SVN et suivi post-thérapeutique : T1, T2 FS/DIXON 3 plans, séquence Écho de Gradient T2* (détection du blooming d'hémosidérine) et 3D T1 FS Gadolinium • Comparaison volumétrique à 6 mois avant consultation multidisciplinaire rhumato-orthopédique.

4. Points Clés à Retenir pour la Pratique

- Signe IRM caractéristique de la TTCG/SVN : hyposignal T1/T2 avec artefact de susceptibilité ('blooming') sur la séquence T2* lié à l'hémosidérine.
- Le suivi IRM à 6 mois sous traitement médical évalue la réduction du volume synovial charnu rehaussé (souvent > 30 % dès 3–6 mois) et la stabilisation des érosions osseuses.
- En cas de masse nodulaire intra-articulaire atypique non hémosidérinique (diagnostic différentiel : chondromatose synoviale, lipome arborescent, sarcome synovial), la biopsie synoviale sous scanner/écho s'impose.
- Pour les contrôles IRM post-cryothérapie (rachis lombaire, cervical, membres à 6 semaines et 3 mois), les séquences T1 FS Gado avec soustraction distinguent le halo inflammatoire périphérique bénin d'un nodule résiduel.

5. Original PubMed Abstract (Texte Intégral de l'Abstract)

Tenosynovial giant cell tumors (TGCT), are rare colony stimulating factor-1(CSF-1)-driven proliferative disorders affecting joints. Diffuse-type TGCT often causes significant morbidity due to local recurrences

necessitating multiple surgeries. Imatinib mesylate (IM) blocks the CSF-1 receptor. This study investigated the long term effects of IM in TGCT. We conducted an international multi-institutional retrospective study to assess the activity of IM: data was collected anonymously from individual patients with locally advanced, recurrent or metastatic TGCT. Sixty-two patients from 12 institutions across Europe, Australia and the United States were identified. Four patients with metastatic TGCT progressed rapidly on IM and were excluded for further analyses. Seventeen of 58 evaluable patients achieved complete response (CR) or partial response (PR). One- and five-year progression-free survival rates were 71% and 48%, respectively. Thirty-eight (66%) patients discontinued IM after a median of 7 (range 1-80) months. Reported adverse events in 45 (78%) patients were among other edema (48%) and fatigue (50%), mostly grade 1-2 (89%). Five patients experienced grade 3-4 toxicities. This study confirms, with additional follow-up, the efficacy of IM in TGCT. In responding cases we confirmed prolonged IM activity on TGCT symptoms even after discontinuation, but with high rates of treatment interruption and additional treatments.

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6. Full-Text Sections (Europe PMC Open Access JATS XML)

Introduction

Tenosynovial giant-cell tumor (TGCT), historically known as pigmented villonodular synovitis (PVNS), is a rare, at times locally aggressive neoplasm affecting the joints or tendon sheaths in young adults. It is most common around large joints such as the knees, ankles and hips1,2. Known subtypes are localized and diffuse TGCT. The localized subtype comprises a single nodule and has a favorable course while the diffuse subtype involves the synovial lining as well as surrounding structures and is associated with a significant risk of recurrence (>50% depending on follow up times), despite being a benign neoplasm2–4. Metastatic forms have been described, but seem to occur very rarely5,6.

Surgical resection is the primary treatment for both subtypes. However, diffuse TGCT is difficult to remove completely and often requires a total synovectomy, or at times a joint replacement, or rarely even amputation1,2,7. In patients with extensive and/or recurrent TGCT, other available treatment modalities include radiation synovectomy8, external beam radiation therapy9, and cryosurgery10. Their therapeutic value has only been assessed in retrospective, in most cases single center series and their long term side effects and complications are poorly described11.

Recurrent TGCT is rarely lethal, but frequently becomes a debilitating chronic illness with substantial morbidity to the joints and quality of life impairment, caused by the disease itself and the multiple treatments2,12.

In TGCT, a neoplastic clone constitutes a subpopulation (2–16%)13 of cells that overexpress colony-stimulating factor-1 (CSF-1). A t(1;2) translocation that links the CSF1 gene on chromosome 1p13 to the COL6A3 gene on chromosome 2q35 has been described and is believed to be responsible for the overproduction of CSF1 by neoplastic cells13,14. Inhibition of CSF1/CSF-1 receptor (CSF-1R) signaling has shown efficacy in the treatment of locally advanced and recurrent diffuse TGCT15–17.

Imatinib mesylate (IM) inhibits the CSF-1R kinase among other kinases17. It has been shown that inhibition of CSF-1R by imatinib is competitive with ATP, with a Ki value of 120 nmol/L18. We have previously reported on the efficacy of IM in TGCT. In the present study we provide long term follow-up on these initial patients and data on 33 additional consecutive patients.

Methods

This retrospective study was conducted at 12 referral centers across Europe (9 institutions), the United States of America (2 institutions), and Australia (1 institution). The file of all patients with locally advanced, recurrent or metastatic TGCT, treated with IM were reviewed. Patient information was extracted from individual patients' files at each institution by the local investigators and was provided in an anonymous form for final analyses. Histopathologic examination was performed at center of origin by pathologists with extensive experience in mesenchymal tumors. Response was measured using version 1.0 of Response Evaluation Criteria in Solid Tumors (RECIST). Data were described using percentages for qualitative variables and medians with ranges

for continuous variables.

Patients were not treated following a fixed regimen. The study protocol and retrospective analysis was approved by the Ethics Committee in Lyon (Committee for the Protection of Individuals, Sud-Est IV, Lyon, France – L10-153 dated 9 December 2010) and was carried out in accordance with the applicable rules concerning the review of research ethics committees. Patients provided written informed consent to treatment with 'off-label' medication, for research review and analysis of medical records. Treatment decision was left to the treating physician. The study was conducted in accordance with ethical requirements that differed per country. National investigators dealt with it according to standard practice. All 12 centers at which the study was carried out approved access to the data.

Survival was plotted using the Kaplan-Meier method. Progression-free survival (PFS) was calculated from the date IM was started to the date of disease progression or death. The time to treatment failure (TTF) was calculated from the date IM was started to the date it was stopped because of toxicity, disease progression, or death, whichever occurred first. For patients with a surgical resection or other additional therapy after treatment with IM, PFS and TTF were censored at the time of surgery. Disease specific survival was calculated from the date IM was started to the date of death due to TGCT. Symptomatic response was defined as improvement of pain and/or joint function in patients who had symptoms at baseline. All statistical analyses were performed using R version 3.4.0 (R Foundation, Vienna, Austria) with packages 'ggplot2', 'rms', and 'survival'.

Results

A total of 62 patients with histopathologically proven TGCT treated with imatinib were identified, their main characteristics are described in Table 1. Briefly, median age at diagnosis was 39 (interquartile range (IQR) 31–53) years and 45 (IQR 36–56) years at start of treatment with IM, the majority of patients were female (N = 39, 63%), and the knee (N = 35, 56%) was the most commonly affected joint (Table 1). At start of IM treatment, three (5%) patients had biopsy proven metastatic disease, 15 (24%) locally advanced disease and 44 (71%) locally recurrent disease. Among patients with prior operations for TGCT (n = 47), the median number of prior operations was 2 (range 1–9), and the time since the last operation was 23 (range 1–192) months. Median follow up of all patients was 52 (IQR 18–83) months.Table 1Descriptive of diffuse-type TGCT patients receiving imatinib mesylate treatment.Patients N (%)Total62 (100)Median age at diagnosis (IQR), yrs.39 (31–53)Median time from diagnosis to start IM (IQR), yrs.3.5 (1–8) Sex Male23 (37)Female39 (63) Tumor location Knee35 (56)Ankle11 (18)Hip6 (10)Foot4 (6)Shoulder1 (2)Elbow1 (2)Head and Neck2 (3)Wrist2 (3) Surgery before start IM None15 (24)1–224 (39)3–413 (21)>410 (16)Median N of surgeries (range)2 (1–9)Median time since last surgery (range), mo.23 (1–192) Disease status Locally advanced20 (32)Recurrence after surgery39 (63)*Metastatic disease3 (5)Abbreviations: TGCT = Tenosynovial Giant Cell tumor, IM = imatinib mesylate, N = Number of patients, mo = months, yrs = years. *One of the locally recurrent patients progressed to metastatic disease.

Descriptive of diffuse-type TGCT patients receiving imatinib mesylate treatment.

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Sixty-one patients received 400 mg and one patient received 600 mg IM daily, all as a single dose/day. The 3 patients with metastatic disease at treatment start progressed rapidly on IM and were excluded from further analysis. One other patient with metastatic disease after multiple surgical treatments and IM, was excluded for further analyses too, leaving 58 patients for the rest of the analysis.

Median duration of IM treatment was 9 (IQR 5–27) months. At last follow-up, the majority of patients (n = 38; 66%) had discontinued treatment. Seventy-seven percent (95% CI 67–89), 41% (95% CI 29–57) and 36% (95% CI 25–52) of patients were still on IM after 6-, 12- and 24-months, respectively (Fig. 1). The treatment failure-rate was 18% (95% CI 6–29) after 12 months.Figure 1Kaplan–Meier survival curves showing the duration of imatinib mesylate treatment (yellow line) and progression free survival (PFS; blue line) in patients with locally advanced or recurrent diffuse-type TGCT. PFS was calculated from the date imatinib mesylate was

started to the date of disease progression or death. The shaded areas are 95% confidence intervals (CI). Over half of the patients discontinued IM within a year. The overall PFS after 5 years was ~50%.

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Response could not be assessed in 3 patients, two of which were lost to follow-up and one who discontinued early due to febrile neutropenia, leaving 55 patients with locally advanced or locally recurrent TGCT assessable for response. Seventeen patients (31%; 95% CI 19–43) had a RECIST-defined response, including 2 (3%) patients with a complete response. The median time to best response was 6 (range 1–23) months.

Forty of 51 patients (78%) reported symptom improvement (Table 2), including 14 of 15 patients with radiological response (CR or PR). Among patients with radiological stable disease (SD), 22 of 30 patients (73%), for whom data was available, had symptom improvement.Table 2Summary of imatinib mesylate efficacy in patients with locally advanced or recurrent diffuse-type TGCT.ParameterPatients N (%) RECIST best response* Complete remission2 (4)Partial response15 (27)Stable disease36 (65)Progressive disease2 (4)Overall response rate17 (31)Rate of disease control53 (96)Symptomatic response40 (78)**Median IM treatment duration (IQR), mo.9.3 (5–26)Median PFS (IQR), mo.18 (8–55)Abbreviations: TGCT = Tenosynovial Giant Cell tumour, IM = imatinib mesylate, N = Number of patients, mo = months, yrs = years, IQR = inter quartile range.Overall response rate includes complete remission and partial response; Rate of disease control includes complete remission, partial response and stable disease; Symptomatic response was indicated as present or not (40/51 = 78%). Metastatic patients (n = 4) were excluded.*N = 3 RECIST best response not available; **N = 9 symptomatic response not available.

Discussion

To our knowledge, this retrospective study provides the largest case series, with long follow-up, of patients with locally advanced, recurrent or metastatic diffuse-type TGCT treated with IM. We confirmed that IM has activity in TGCT with an overall response rate of 31% in patients with locally advanced/recurrent TGCT. Interestingly all patients with metastatic TGCT progressed on IM, suggesting that metastatic TGCT is either a different disease or loses its dependency on the CSF1/CSF1R axis during malignant transformation. The main issue, is the drop-off rate, with more than half of the patients discontinuing therapy within a year of therapy (59%; 95% CI 29–57), in most cases for unclear reasons (patients decision, physician’s decision) suggesting an unfavorable efficacy/toxicity balance. Eleven percent of patients reported grade 3–4 toxicities, which is consistent with the rates reported with IM for adjuvant gastrointestinal stromal tumors (GIST) or chronic myeloid leukemia (CML)19–22.

To date, surgical resection remains the treatment of choice for diffuse-type TGCT, but is associated with high recurrence rates and multiple additional surgeries11. It is challenging to balance between increased morbidity of multiples or invasive surgeries12,23, alternative therapeutic options, and daily symptoms of the tumor. A more aggressive resection or other multimodality treatments, such as external beam radiation therapy, radiosynovectomy and cryosurgery, may adversely affect joint function, quality of life and development of osteoarthrosis, which, given the young age group, are relevant factors2,24. This would justify a less invasive approach, using systemic therapy, provided those are associated with tumor shrinkage and, most importantly, symptomatic improvements25.

In the present study, age, localization and gender distribution were consistent with the literature10,24,26. The extent of disease in our patient group is emphasized by a disease specific survival of 90% including four metastatic patients and 49% of patients had three or more surgeries before start IM. Similar to previous case-series, we calculated a 1- and 5-years PFS of 71% and 48%, metastatic patients excluded, respectively10,24,26. Because of heterogeneity of patients and a variety of treatments, it is debatable to compare these numbers.

The overall response rate appears higher compared to nilotinib 6% (95% CI unknown), a different tyrosine kinase inhibitor, with similar potency against CSF1R27. Our overall response rate 31% (95% CI 19–43, metastatic patients (N = 4) excluded) was consistent with our previous report on the short term results of IM 19% (95% CI 4–34) with similar disease control rate (96% versus 93%)17. In the present study, 38 (66%) patients discontinued IM; 14 (37%) without subsequent treatment, of which ten patients had stable disease at final follow up. Of the 38 patients who discontinued IM, 21 patients (55%) discontinued IM for toxicity or non-specific medical reasons. 13 (62%) out of these 21 patients eventually progressed. Both stable and progressive patients can be a result of discontinuing IM treatment or the natural course of disease.

Newer, more specific inhibitors of CSF1R, currently only available in trial-setting such as emactuzumab (RG7155)28, pexidartinib (PLX3397)15,29,30, and cabiralizumab31 (FPA008, Five-Prime), have shown promising clinical activity on similar groups of diffuse TGCT patients in prospective clinical studies with more formal criteria and timelines for response assessment than this retrospective series. Emactuzumab (N = 29)16 had an overall response rate of 86% (two patients with a complete response) and a rate of disease control of 96%, including a significant functional and symptomatic improvement (median follow up 12 months). Pexidartinib showed (N = 23)15 an overall response rate of 52% (all patients had a partial response) and a rate of disease control of 83%. At ASCO 2018 results of a pexidartinib placebo controlled, phase 3 study showed a significant improved overall response rate (39.3% vs 0%) and PROMIS physical function (4.06 vs 0.89), after a median 6 months follow up29. The preliminary results with cabiralizumab (N = 22) are consistent, with radiographic response and improvement in pain and function in five out of 11 patients31. However, long term efficacy data have not yet been reported with these newer agents.

Virtually all patients treated with IM for either CML or GISTs, experience32 at least one mild or moderate adverse effect (grade 1–2). Toxicities of IM are determined by the disease stage and the doses used, advanced disease and higher doses result in more frequent and severe toxicities. Most side effects occur early in the course of treatment and tend to decrease in frequency and intensity in time32. We consider a 10–15% rate of grade 3–4 toxicities in a generally benign but locally aggressive disease, such as diffuse TGCT, too high. Only 22% of patients did not experience any side effects.

Although target anti-cancer therapies are described as ‘well tolerated’, the perception of tolerability may vary in the context of a, most often, benign condition. Understanding, monitoring and managing the side effects will be important to optimize systemic therapy for patients with TGCT.

Discontinuation of treatment due to toxicities was seen for IM (this series), emactuzumab15 and pexidartinib16 in 12%, 20% and 9% patients, respectively. TGCT patients might be less willing to cope with adverse event-related and study-related procedures. Here, we report prolonged clinical benefit and symptomatic relief, even after discontinuation of treatment. A similarly persistent effect was observed with monoclonal antibodies and more specific CSF1R tyrosine kinase inhibitors25. This suggest that intermittent treatment administration may be an option to improve long term tolerability.

Conclusion

Identification of a biologic aggressive subgroup of diffuse TGCT, at risk of increased surgical morbidity or recurrent disease, should aid to decide which patients benefit most of systemic treatments. With the advent of more potent CSF-1R inhibitors, such as emactuzumab, pexidartinib and cabiralizumab, the role of IM in extensive TGCT might weaken, but may be balanced by the favorable safety profile of IM. Availability of these new compounds, both in terms of registration and reimbursement, will ultimately define the prescribed drug in daily practice37.